

COMMUNICATION PROTOCOL

UAS MINI PROJECT

Credit Score Mock API

Enterprise Credit Scoring REST API dengan Observability Dashboard — mensimulasikan proses credit scoring menggunakan REST API dengan fitur audit log, Request ID, Correlation ID, response time, dan status code.

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Universitas Cakrawala 2026

Latar Belakang

Problem Statement

Lembaga keuangan membutuhkan sistem yang mampu menilai risiko calon peminjam secara cepat, konsisten, dan terdokumentasi — menerima data pengajuan, melakukan penilaian, menghasilkan keputusan kredit, serta menyediakan mekanisme monitoring terhadap seluruh aktivitas API.

REST API

Membangun REST API sederhana menggunakan TanStack Start

HTTP + JSON

Implementasi komunikasi HTTP berbasis JSON

Postman

Pengujian endpoint menggunakan Postman

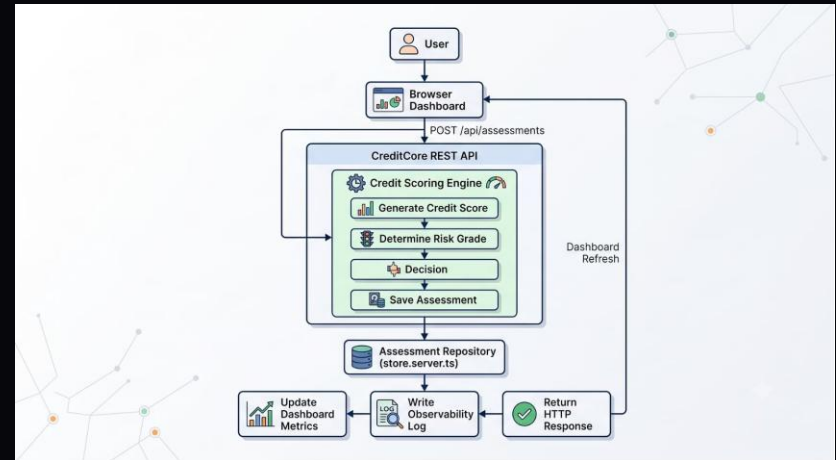
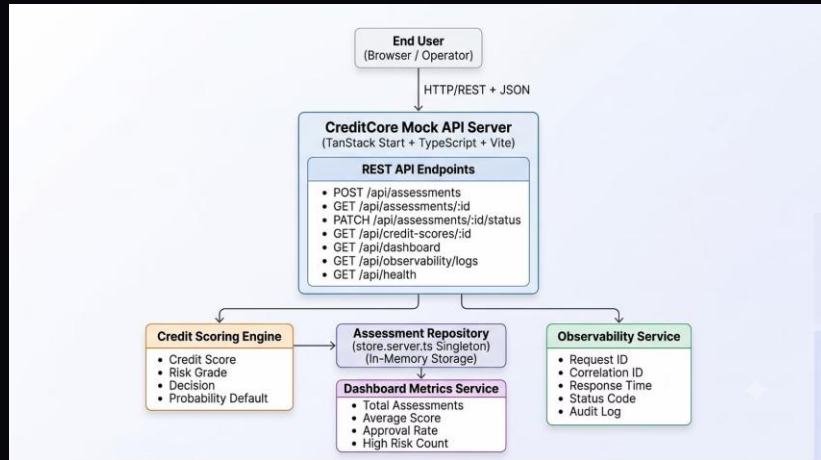
Observability

Monitoring terhadap seluruh request API

Use Case: Model Scoring Mock API

Arsitektur Sistem

Sistem terdiri dari End User, REST API Server, Credit Scoring Engine, Assessment Repository, Dashboard Metrics Service, dan Observability Service. Client mengirim request via HTTP/REST, server memproses melalui Credit Scoring Engine, menyimpan hasil ke repository, dan dashboard menampilkan statistik.



Endpoint Design

Business & Infrastructure API

Method	Endpoint	Fungsi
POST	<code>/api/assessments</code>	Membuat assessment baru
GET	<code>/api/assessments/:id</code>	Detail assessment berdasarkan id
PATCH	<code>/api/assessments/:id/status</code>	Update status assessment
GET	<code>/api/credit-scores/:id</code>	Melihat credit score
GET	<code>/api/health</code>	Memeriksa status server
GET	<code>/api/dashboard</code>	Menampilkan statistik dashboard
GET	<code>/api/observability/logs</code>	Menampilkan audit log

Postman Collection juga menyediakan endpoint simulasi (`GET/PUT /api/simulation`) untuk menguji skenario error seperti HTTP 500 tanpa mengubah kode aplikasi.

Success Scenario

Response Berhasil

POST /api/assessments — 201 Created

The screenshot shows a REST client interface with a workspace titled "M. Maulana Kherul Anang's Workspace". The active collection is "CreditCore - Mock API" and the selected endpoint is "POST /api/assessments". The request body is a JSON object with the following structure:

```
{
  "success": true,
  "data": {
    "assessmentId": "A00-2020-0015",
    "status": "Pending",
    "createdAt": "2020-07-12T06:32:15.570Z",
    "updatedAt": "/api/assessments/A00-2020-0015",
    "scoreAssessment": "/api/assessments/A00-2020-0015"
  },
  "error": null,
  "meta": {
    "requestId": "2020-07-12T06:32:15.570Z",
    "correlationId": "2020-07-12T06:32:15.570Z",
    "timestamp": "2020-07-12T06:32:15.570Z",
    "latencyMs": 0,
    "path": "/api/assessments",
    "method": "POST",
    "source": "Postman"
  }
}
```

The response is a 201 Created status with a 5 ms response time and a 1.02 KB body. The response body is a JSON object with the following structure:

```
{
  "success": true,
  "data": {
    "assessmentId": "A00-2020-0015",
    "status": "Pending",
    "createdAt": "2020-07-12T06:32:15.570Z",
    "updatedAt": "/api/assessments/A00-2020-0015",
    "scoreAssessment": "/api/assessments/A00-2020-0015"
  },
  "error": null,
  "meta": {
    "requestId": "2020-07-12T06:32:15.570Z",
    "correlationId": "2020-07-12T06:32:15.570Z",
    "timestamp": "2020-07-12T06:32:15.570Z",
    "latencyMs": 0,
    "path": "/api/assessments",
    "method": "POST",
    "source": "Postman"
  }
}
```

GET /api/dashboard — 200 OK

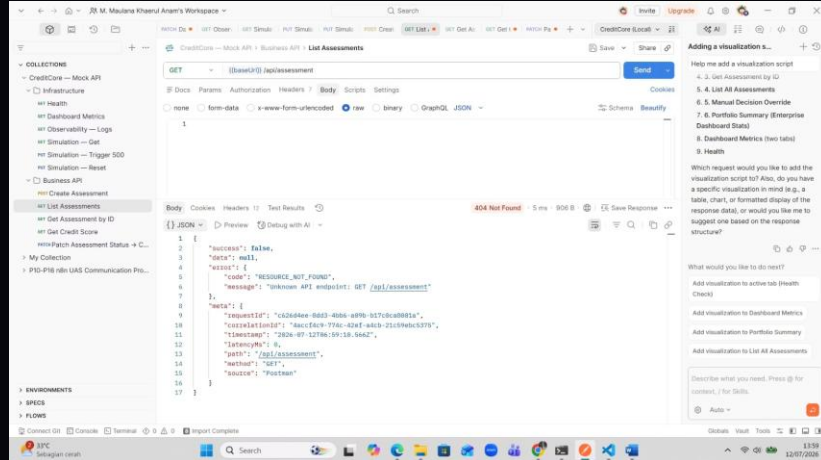
The screenshot shows a REST client interface with a workspace titled "M. Maulana Kherul Anang's Workspace". The active collection is "CreditCore - Mock API" and the selected endpoint is "GET /api/dashboard". The response is a 200 OK status with a 5 ms response time and a 4.84 KB body. The response body is a JSON object with the following structure:

```
{
  "success": true,
  "data": {
    "totalAssessments": 21,
    "averageCreditScore": 732,
    "approvedRate": 70,
    "averageRequestAmount": 1871905,
    "averageMonthlyIncome": 14842007,
    "highestAssessments": 2,
    "totalAssessments": 180,
    "completedCases": 21,
    "scoreAssessments": [
      {
        "account": "A00-2020-0002",
        "scoreAssessment": "New Setpoint",
        "requestAmount": 18000000,
        "monthlyIncome": 1000000,
        "employmentLengthMonths": 24,
        "postponedLoan": "Business Ventures"
      }
    ]
  }
}
```

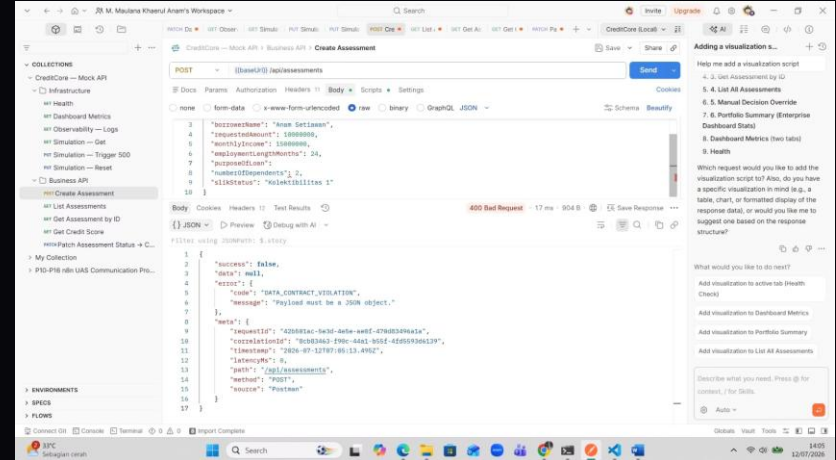
Failure Scenario

Response Kegagalan

Get /api/assessments — 404 Not Found



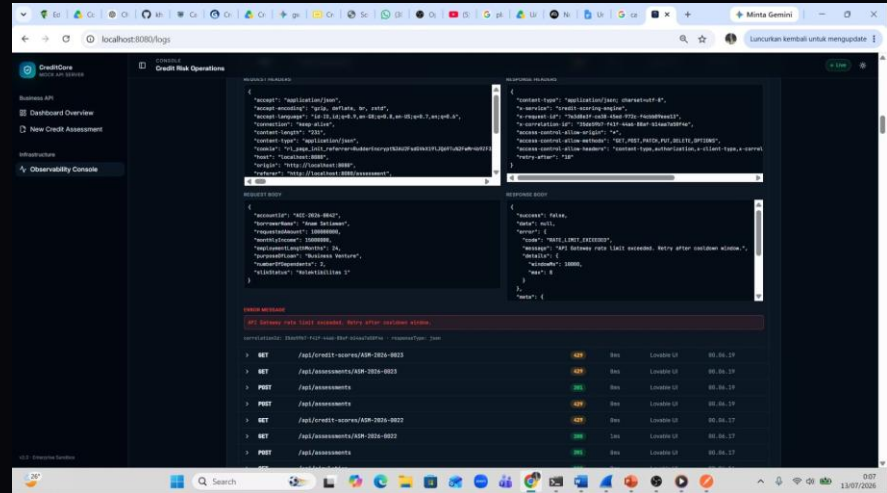
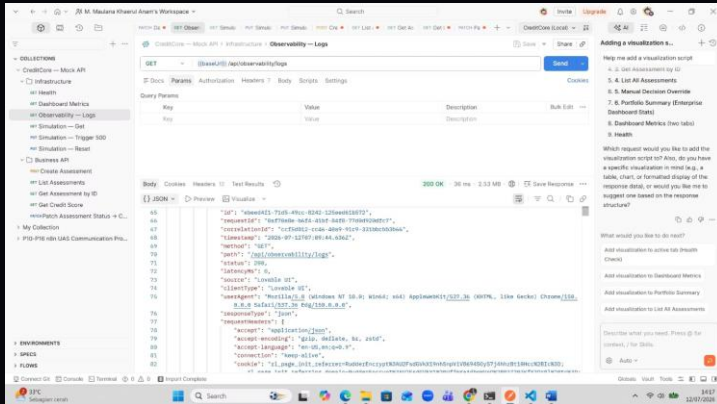
Post /api/assessments — 400 Bad Request



Observability Evidence

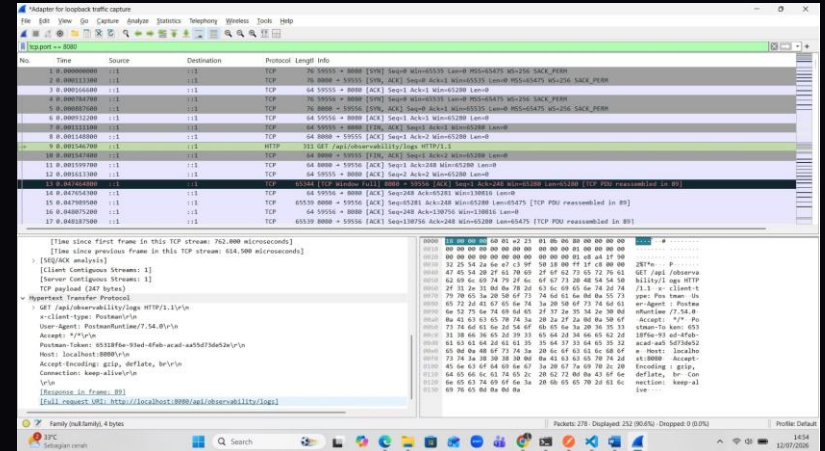
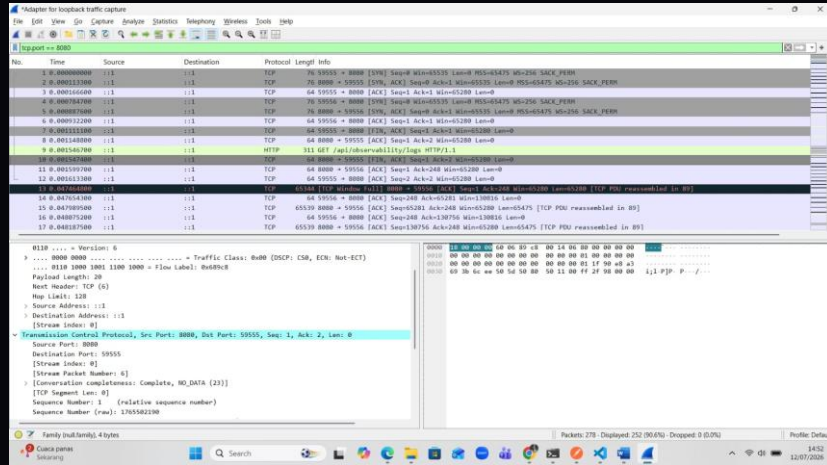
Audit Log & Monitoring

Sistem menyediakan observability melalui endpoint GET /api/observability/logs. Setiap request mencatat informasi lengkap untuk monitoring dan debugging real-time.



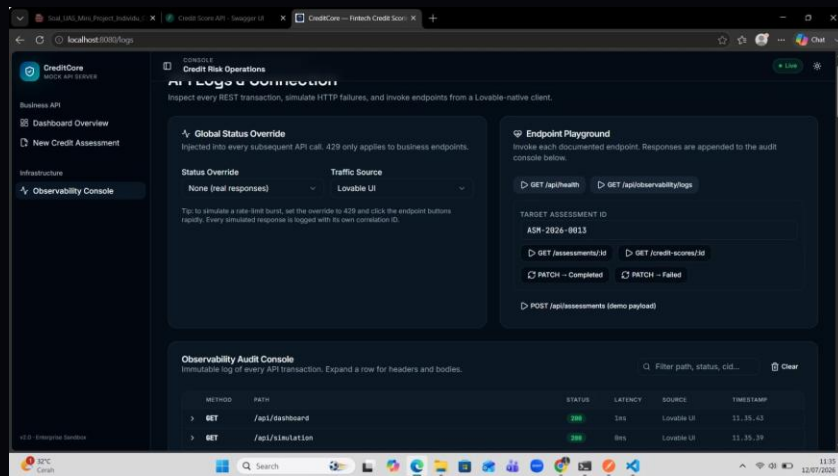
Analisis Paket Jaringan

loopback interface dengan source dan destination sama-sama ::1.



Demo App

Pada demo ini ditampilkan aplikasi Credit Score Hub yang berfungsi sebagai mock credit scoring platform. Dashboard mengambil data langsung dari backend melalui REST API.



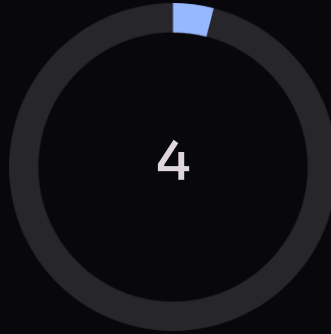
Testing Result & Conclusion

Hasil Pengujian & Refleksi



Pengujian Lulus

Create, Get, Dashboard, Credit Score, Observability, Health, Error Testing



Status Code

200 OK, 201 Created, 400 Bad Request, 404 Not Found



Reliability

Response sesuai kontrak API dengan audit log lengkap

Limitation

- In-Memory Storage, data hilang saat server berhenti
- Belum menggunakan database PostgreSQL/MySQL
- Belum menerapkan autentikasi & otorisasi
- Belum menggunakan HTTPS

Improvement

- Integrasi database PostgreSQL
- Autentikasi JWT
- Docker untuk deployment
- Workflow otomatisasi n8n
- Rate limiting & monitoring komprehensif

GitHub: github.com/khaerulanamm/creditscorehub • **Video Demo:** <https://youtu.be/DgVX7of2MD0>